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Photophysical studies of coordination polymers and composites based on heterometallic lanthanide succinate

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This work reports the synthesis and characterization of heterometallic Tb/Eu lanthanide-succinate coordination polymers (CPs) under hydrothermal conditions. Compounds were obtained inserting different quantities of Eu^{3+} ions into the terbium succinate structure, named here **xEuyTbSuc** (x and y are the percentage of the respective lanthanide ions). Composites with these heterometallic CPs and activated carbon (CP@AC, AC = activated carbon) were obtained using one-pot hydrothermal reactions. Bimetallic CPs with or without presence of AC crystallize in monoclinic system, isostructural to $[Ln_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$ (Ln = Eu or Tb), already described in the literature, confirmed by X-ray powder diffraction (XRPD) data. Scanning electron microscopy (SEM) images show the CP crystals inside the carbon pores. Luminescence spectroscopy measurements (excitation, emission and lifetime) were conducted to explore photophysical properties of these new materials. Lifetime data indicates all Eu^{3+} ions are in the same chemical environment. Even at low concentration (1%), Eu^{3+} transitions ($^5D_0 \rightarrow {}^7F_J$ (J = 0–4)) were observed in the emission spectrum, along transitions related to the Tb^{3+} cations ($^5D_4 \rightarrow {}^7F_J$ (J = 6–3)), due to the metal-to-metal charge transfer (MMCT) between the lanthanide ions. Similar results are found in the composites. Increasing the Eu^{3+} concentration, a more intense MMCT process took place, suppressing the terbium emission.

1. Introduction

In the scope of coordination polymers (CPs), flexible ligands have gain great attention due to their structural diversity related to different conformations and coordination modes adopted by these organic molecules. They are directly associated to the control of experimental parameters such as pH, temperature, time and so on [1]. Within this class of ligands, succinic acid and its derived have been often explore to produce CP-structures with wide variety of connectivity and topologies, especially with lanthanide trivalent ions [2]. For many years, studies about lanthanide succinate-derived coordination polymers were mainly focused on the structural characterization, however, many works have reported other properties of these materials, for example, catalytic activity in some organic reactions [3–5]. In recent years, luminescent properties of Ln-CPs based on succinic acid derived suchlike methylsuccinic acid, 2,2-dimethylsuccinic acid, 2,3-dimethylsuccinic acid and phenylsuccinic acid were explored, showing potential applications of these materials in chemical sensing [6–11].

One way to enhanced and explore new properties in material science is obtain composites [12,13]. Several carbonaceous materials were used to synthesize CP-based composites, as quantum dots, carbon nanofibers, graphene oxide, etc. [13]. These materials are applied in several fields of research such as adsorption, chemical sensing and catalysis [13]. Our research group has been investigated the obtention of Ln-CPs composites based on activated carbon (AC) and, as well, their structural and morphological characterization [14–16]. These composites were synthesized using one-pot hydrothermal reactions, crystallizing lanthanide coordination polymers inside the activated carbon pores (CP@AC), confirmed by scanning electron microscopy [14–16]. We also

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demonstrated the applicability of these carbon-based composites on the selective separation of dyes in aqueous solution [15]. For lanthanide succinates (LnSuc = $[Ln_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$) coordination networks, several composites LnSuc@AC (Ln = Pr, Nd, Sm, Eu, Gd, Tb, Dy, Er and Tm, AC = activated carbon) [14,16]. The system based on **TbSuc** shows detoxifying properties for pesticides in biological systems [16].

Thus, we explored the luminescent properties of hetero-bimetallic systems based on lanthanide succinates and their respective **LnSuc@AC** composites (Ln = Eu and Tb, Suc = succinate). The bimetallic CPs and composites were obtained using hydrothermal, starting of several molar ratios of europium and terbium ions. In this way, we insert different portions of Eu^{3+} (1%, 10%, 25% and 50%) into the **TbSuc** structure. The materials were characterized by Scanning Electron Microscopy (SEM), confirming the typical morphology of CP@AC composites, already reported in the literature [14–16]. Crystalline phases inside the carbon pores were identified by X-ray powder diffraction (XRPD). Photophysical properties in presence and absence of activated carbon, and the influence of molar proportion in the crystalline structure were investigated using luminescence spectroscopy in solid state at room temperature. Excitation spectra, emission spectra and lifetime measurements are in agreement with XRPD data.

2. Materials and Methods

2.1. Materials

Succinic acid ($C_4H_6O_4$, 99.5%) and Ln_2O_3 (Ln = Eu and Tb, 99.9%) were obtained from Sigma-Aldrich. The activated carbon (AC) was obtained from the Dinâmica Química. All reagents were used without previous purification. Lanthanide chlorides $LnCl_3\bullet 6H_2O$ (Ln = Eu and Tb) were synthesized by reaction of hydrochloric acid with the respective lanthanide oxide.

2.2. Experimental procedure

2.2.1. Synthesis of $[Ln_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$ (LnSuc; Ln = Eu or Tb; Suc = succinate)

Lanthanide succinates were synthesized using hydrothermal reaction, as already reported in the literature [14]. In a Teflon reactor (23 mL), 0.5 mmol (0.059 g) of succinic acid was added in 10 mL of deionized water. The pH was adjusted to 5–6 using NaOH solution, follow by the addition of the corresponding lanthanide chloride (0.5 mmol). The system was sealed and heated until 120 °C. This temperature was hold for 96 h and the reaction system was cooled naturally to room temperature. Colourless crystals obtained were washed with water and ethanol, and dried at room temperature.

2.2.2. Synthesis of heterometallic succinates (1Eu99TbSuc, 10Eu90TbSuc, 25Eu75TbSuc and 50Eu50TbSuc)

Mixed metal succinate coordination polymers were obtained using the same procedure above, but starting of different molar ration of europium and terbium chlorides. Heterometallic lanthanide succinates were obtained using the Eu/Tb molar ratio: 0.005/0.495 (for **1Eu99TbSuc**), 0.05/0.45 (for **10Eu90TbSuc**), 0.125/0.375 (for **25Eu75TbSuc**) and 0.25 of each lanthanide chloride (for **50Eu50TbSuc**). Crystals obtained were washed with water and ethanol, and dried at room temperature.

2.2.3. Synthesis of heterometallic EuTbSuc@AC composites (1Eu99TbSuc@AC, 10Eu90TbSuc@AC; 25Eu75TbSuc@AC and 50Eu50TbSuc@AC)

Composites based on heterometallic Eu/Tb succinates were obtained using the same procedure of their respective coordination polymers, except by the addition of activated carbon in the reaction mixture (50% in the sum of the starting reagents masses).

2.3. Physical measurements

Powder diffraction patterns were recorded in a Shimadzu diffractometer XRD-700 with Kα(Cu) 1.54 Å source, step 0.01°, acquisition time of 1 s and window 5–50°. The SEM images were obtained in a Shimadzu SS550 microscope with tungsten filament, working at 20 kV and WD 9.8–10 mm, coupled with an EDS module. Photoluminescence measurements were performed in a modular spectrofluorometer Horiba Jobin-Yvon Fluorolog-3 with double monochromator, using a 450 W xenon lamp.

3. Results and discussion

Some europium and terbium succinates coordination polymers had been synthesized in the literature under different reaction conditions of pH, molar ratio and temperature [2,18,19]. Although succinic-based heterometallic systems are in evidence in recent years [8–10], Eu/Tb mixed-metals CPs with this ligand were not reported yet.

In this work, hydrothermal reactions had been used to obtain heterometallic coordination polymers and their carbon composites based on Eu^{3+} and Tb^{3+} cations. All bimetallic CPs crystallize in monoclinic system and I2/a space group, and are isostructural to $[Ln_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$ (Ln = Eu or Tb), well described in the literature [17,18]. Experimental XRPD are shown in Fig. 1, and confirm the crystalline phases. The absence of additional peaks indicates that no other solid phase was obtained. Therefore, the isomorphic substitution causes no significant distortions in the crystalline structure, since the lanthanide cations have similar ionic radii. In all diffraction patterns, the most intense peaks are located around 10.5°, related to the (002) diffraction plane.

CP@AC composites were obtained also via hydrothermal reactions, by the addition of activated carbon in the reaction solution. Fig. 2 show the XRPD patterns of composites containing lanthanide succinate with different molar ratio of Eu^{3+} and Tb^{3+} in their

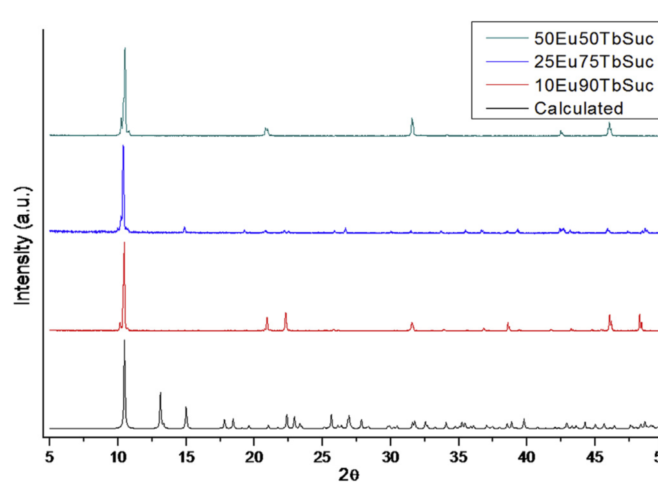

Fig. 1. Experimental powder patterns of the Eu/Tb succinates, in comparison to the calculated from the $[Tb_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$ [18].

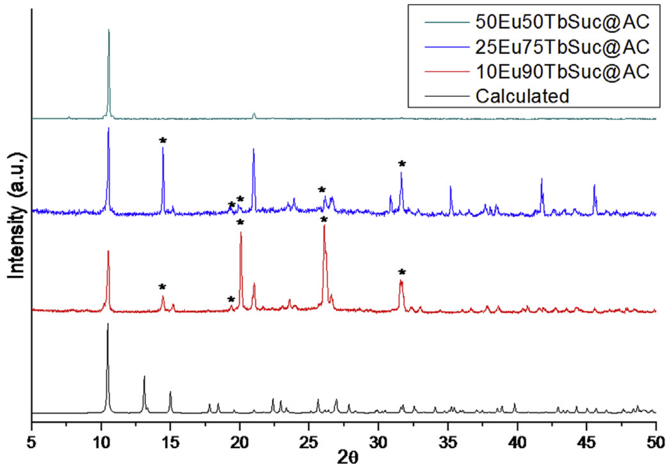

Fig. 2. Experimental powder patterns of composites in comparison to the $[Tb_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$. (*impurities).

structure. Similar signals were observed in comparison to the AC free reactions, indicating same monoclinic structure crystallized inside the pores of the carbon matrix. Additional peaks were observed for the composites **10Eu90TbSuc@AC** and **25Eu75TbSuc@AC**, highlighted in Fig. 2. These signals are related to the carbon matrix [16] and impurity of unreacted succinic acid [19,20]. In all powder patterns the most intense signal around 10.5° corresponding to the (002) diffraction plane.

The composite with equal quantities of Eu^{3+} and Tb^{3+} (**50Eu50TbSuc@AC)** cations was analysed by SEM-EDS and the images are shown in Fig. 3, in comparison to the activated carbon (Fig. 3a). Results confirm the crystallization of LnSuc coordination polymer inside the carbon pores. The morphology observed is in agreement with others CP@AC composites obtained through similar experimental conditions [14–16]. EDS analysis inside the composite pores (Fig. 3d) shows the presence of the lanthanide ions.

Solid state photoluminescence measurements for **TbSuc** and **EuSuc** were performed at room temperature, and the results are shown in Figs. S1 and S2. All signal observed corresponding to 4f-transitions of the respective lanthanide trivalent ions, in agreement with the literature [8–10,16]. Emission spectra show intense peaks at 544 nm and 616 nm, for **TbSuc** and **EuSuc** respectively. These signals are related to $^5D_4 \rightarrow {}^7F_5$ and $^5D_0 \rightarrow {}^7F_2$ transitions, characteristic of the greenish and reddish luminescence of the **TbSuc** and **EuSuc**, respectively. For the **EuSuc** compound, the luminescence lifetime decay curve was obtained at room temperature, monitoring in the $^5D_0 \rightarrow {}^7F_2$ emission (Fig. S3). The single exponential behaviour indicates all europium ions are in the same coordination environment [21], in according to the crystalline structure found in the x-ray powder diffraction results.

Table 1 shows some experimental photoluminescence parameters for **EuSuc** in comparison to similar structures in the literature. The experimental lifetime (1.74 ms) and quantum efficiency (62.5%) are higher than other similar compounds, suchlike $[Eu_2(msuc)_3(H_2O)_2]$ (msuc = methylsuccinate), $[Eu_2(2,3\text{-}dms)_3(H_2O)_2]$ (2,3-dms = 2,2-dimethylsuccinate) and $[Eu_2(psa)_3(H_2O)]$ (psa = 2-phenylsuccinate) [7–9,11], indicating fewer luminescence-quenching channels in the EuSuc.

Since all structures in Table 1 have coordination water molecules in their structure, the additional C–H groups in side chains of methylsuccinate, 2-phenylsuccinate and 2,2-dimethylsuccinate play an important role in the luminescence quenching

Table 1
Radiative (A_{rad}), nonradiative (A_{nrad}) constant rates, experimental lifetime luminescence and quantum efficiency of the EuSuc in comparison to the other similar structures in the literature [7,9,11].

	A_{rad} (s^{-1})	A_{nrad} (s^{-1})	τ (ms)	η (%)
$[Eu_2(Suc)_3(H_2O)_2]\bullet 0.5H_2O$	359.5	216	1.740	62.5
$[Eu_2(msuc)_3(H_2O)_2]$	337.8	1194	0.653	22.1
$[Eu_2(2,3\text{-}dms)_3(H_2O)_2]$	379	1562	0.515	19.5
$[Eu_2(psa)_3(H_2O)]$	386	1562	0.587	22.6

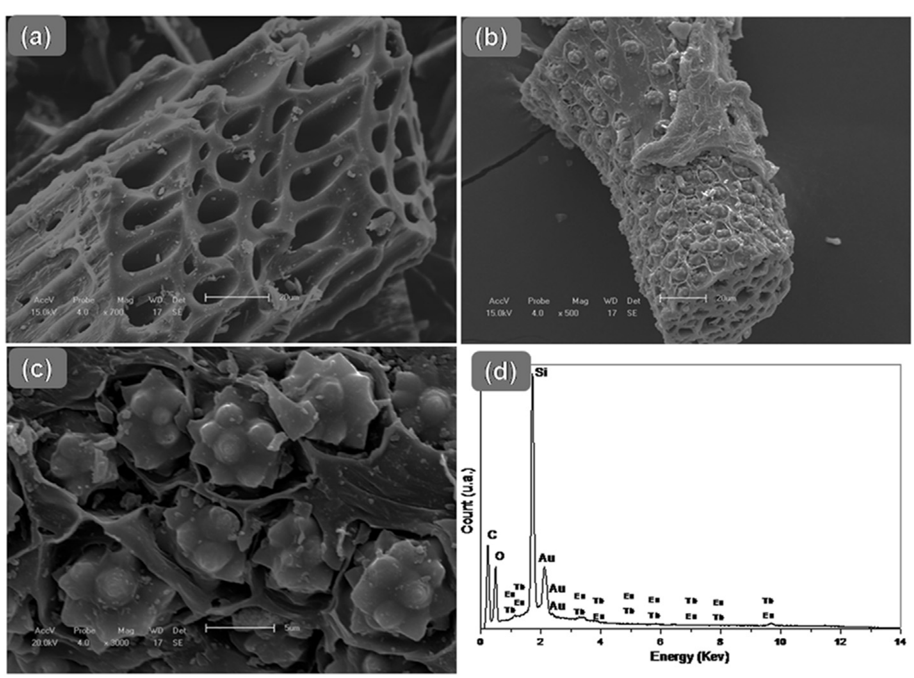

Fig. 3. SEM images of (a) AC, (b–c) **50Eu50TbSuc@AC** and (d) EDS analysis inside the composite pore.

mechanism [7]. This observation can be evidenced comparing the nonradiative rate constants (Table 1) that reflects the energy loss via nonradiative processes, in this case, mainly vibrational relaxation. Another suppression mechanism often used to explain the luminescence of succinic-based materials is the concentration quenching of luminescence [7,9]. This phenomenon is related to the energy transfer processes between lanthanide cations due to their proximity in the crystal structure, leading to decrease in the luminescence intensity [22,23]. In the **EuSuc** structure, the Eu^{3+} ions are separated by 9.443 Å along *c* direction via *trans*-succinate ions and 6.049 Å in *a* direction via *gauche*-succinates. These atomic distances are near to the ones found in the $[Eu_2(2,3\text{-dms})_3(H_2O)_2]$ crystal structure, for example [7]. In this way, the energy losses through the vibrational relaxation still seems to be a more important parameter to understand the higher intensity luminescence of the **EuSuc**, compared to the other similar compounds.

Several heterometallic lanthanide coordination compounds base on succinate-like ligands have been obtained through hydrothermal reactions [8,10]. Thus, different quantities (1%, 10%, 25% and 50%) of Eu^{3+} ions were successfully inserted in the **TbSuc** structure. The excitation and emission spectra of **1Eu99TbSuc** (1% Eu^{3+}) and **10Eu90TbSuc** (10% Eu^{3+}) were measured at room (Figs. 4 and 5). For the **1Eu99TbSuc**, the excitation spectrum shows the transitions related to the 4f-4f transitions from the ground state $^5F_6$ of the Tb^{3+} ions, overlapping the transitions related to the europium ions. However, the increasing of the Eu^{3+} ions in the crystal structure leads to an increasing in the relative emission intensity of the transitions of these cations, as observed in the excitation spectrum of the **10Eu90TbSuc**. In this case, the most intense signals are located at 379 nm and 395 nm, corresponding to the $^7F_6 \rightarrow (^5G_6, ^5D_3)$ and $^7F_0 \rightarrow {}^5L_6$, of the terbium and europium ions, respectively.

1Eu99TbSuc and **10Eu90TbSuc** were irradiated at 378 nm and the emission spectra were recorded at room temperature (Fig. 5). In both cases, the emission spectrum shows the characteristic transitions $^5D_0 \rightarrow {}^7F_J$ (J = 0–4) and $^5D_4 \rightarrow {}^7F_J$ (J = 6–3), of the europium and terbium trivalent ions, respectively. Even with 1% of Eu^{3+} in the structure, similar profile was observed in the emission of both lanthanide ions and at 10% of Eu^{3+}, transitions related to the Tb^{3+} cations are almost not noticed. Since both lanthanides ions have an excitation peak near 378 nm, these observations indicate an efficient metal-to-metal charge transfer (MMCT) from the Tb^{3+} ions to Eu^{3+} ions. This charge transfer phenomenon is often reported in Eu/Tb compounds [24–26], and also observed Angeles Monge and co-workers in the $[Eu_{0.8}Tb_{1.2}(psa)_3(H_2O)]$ (psa = 2-phenylsuccinate) [5]. The excitation and emission spectra of the other heterometallic compounds are similar to the **EuSuc**, showing only signals related to the europium trivalent ion, as observed in others Eu/Tb systems already studied [24].

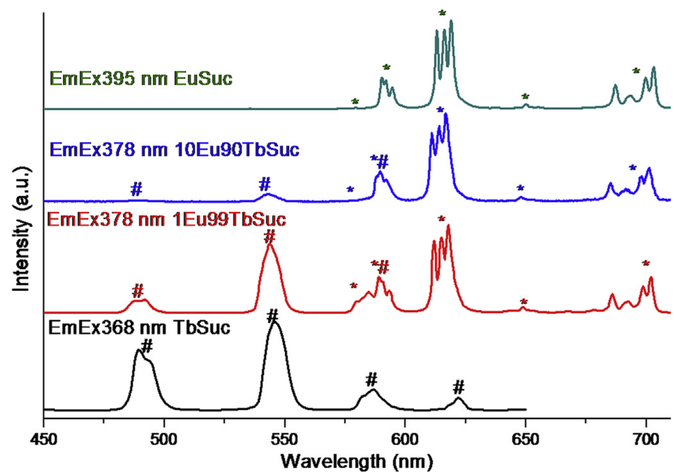

Fig. 5. Emission spectra of **1Eu99TbSuc** and **10Eu90TbSuc** in comparison to the EuSuc and TbSuc. Europium transitions (*) and terbium transitions (#) are also indicated.

Fig. 6 shows the lifetime curves of **1Eu99TbSuc** and **10Eu90TbSuc**, obtained at room temperature, with excitation at 378 nm and monitoring the emission at 617 nm. Both compounds show a single exponential profile, indicating only one Eu^{3+} emission centre, again with agreement with the XRPD data. The decrease in the emission lifetime in the **1Eu99TbSuc** (0.69 ms) reflects the low quantities of europium ion in the structure. The increment in the europium concentration up to 10% leads to higher lifetime (28.6 ms).

Luminescent properties of **LnSuc@AC** composites were also investigated at room temperature. Concerning the different molar ratio in the lanthanide succinates, similar results were found in the excitation and emission spectra. Broader signals were observed in the excitation spectra (Fig. S4) due to the presence of the carbonaceous matrix. Again, the transitions related to the terbium cations are almost not detected in the **25Eu75TbSuc@AC** and **50Eu50TbSuc@AC** composites. Emission spectra were measured at

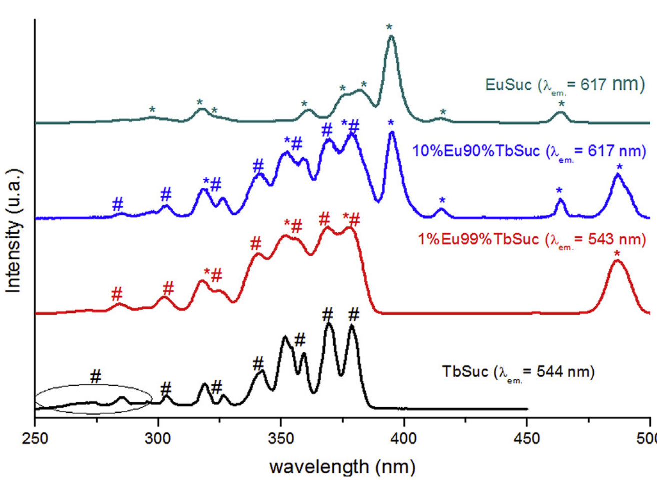

Fig. 4. Excitation spectra of **1Eu99TbSuc** and **10Eu90TbSuc** in comparison to the EuSuc and TbSuc. Europium transitions (*) and terbium transitions (#) are also indicated.

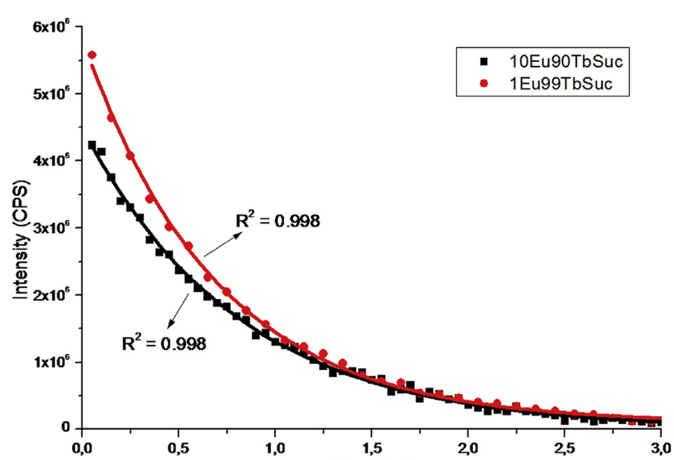

Fig. 6. Lifetime decay curve for **1Eu99TbSuc** and **10Eu90TbSuc** monitoring at 617 nm and excitation in 378 nm.

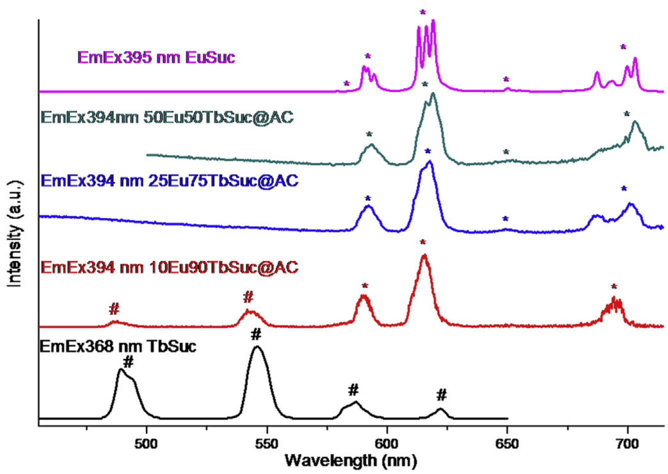

Fig. 7. Solid state emission spectra of the composites at room temperature in comparison to the **TbSuc** and **EuSuc**.

room temperature and shown in Fig. 7. In the **10Eu90TbSuc@AC**, transitions related to both lanthanide ions are noticed, similar to the **10Eu90TbSuc**. However, the ratio between the main transition of Eu^{3+} and Tb^{3+} ions ($I(^5D_0\rightarrow{}^7F_2)/I(^5D_4\rightarrow{}^7F_5)$) reduce from 1.37 to 1.13 in the composite, evincing more intense MMCT when the **10Eu90TbSuc**is inside the carbon material. The emission spectra for **25Eu75TbSuc@AC** and **50Eu50TbSuc@AC** are quite similar to the respective free coordination polymers, as expected, exhibiting only the Eu^{3+} transitions.

Lifetime measurements were done exciting composites at 394 nm, monitoring the hypersensitive transition emission $^5D_0\rightarrow{}^7F_2$ at 617 nm, and the results are in Fig. 8. All experimental decay shows single exponential behaviour, indicating that all europium ions are in the same chemical environment in the composites, in concordance to the experimental X-ray diffraction patterns. The luminescence lifetime (Fig. 8) for the composite with 10% of europium in its structure was 0.34 ms, while value found for the other two composites were 0.84 ms. Smaller lifetime values found in the composites compared to the **EuSuc** due to the reduced quantities of Eu^{3+} cations and C—H bonds of the activated carbon, providing new nonradiative decay channels. It is important to note the increase in the luminescence lifetime in the **25Eu75TbSuc@AC** because of optimization in the metal-to-metal charge transfer between Tb^{3+} and Eu^{3+} cations; however, higher amounts of europium ion in the composite structure did not lead to an improvement in the luminescence decay.

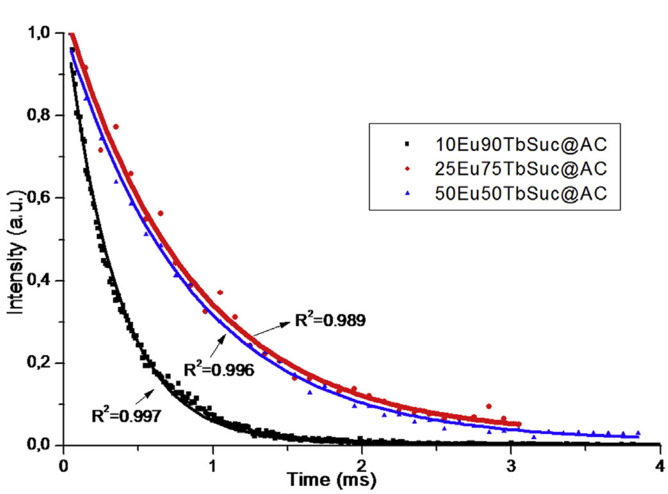

Fig. 8. Luminescence lifetime curves for the composites at room temperature.

4. Conclusion

Here, heterobimetallic Eu/Tb-CPs and its carbonaceous composites were successfully obtained using hydrothermal reactions. The same monoclinic crystalline phase was obtained in absence and presence of activated carbon. Composites present similar morphology to similar systems already reported, with the CP crystals inside the AC pores. Luminescence measurements show the main electronic transitions related to the lanthanide ions and in agreement with the XRPD data. The insertion of Eu^{3+} ions leads to a metal-to-metal charge transfer, causing a decrease in the Tb^{3+} luminescence, even in small amounts of europium cations.

CRediT authorship contribution statement

Guilherme C. Santos: Conceptualization, Methodology, Validation, Formal analysis, Investigation. **Carlos A.F. de Oliveira:** Conceptualization, Supervision, Validation, Formal analysis. **Fausthon F. da Silva:** Conceptualization, Validation, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Supervision.

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Supporting Information

Figs. S1–S4 are available in the supporting information material.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molstruc.2020.127829.

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